

The condition for the block to stay put is governed by two extrema.

**Extremum 1: the block does not move even without  $F$ .** The friction just balances the down-slope component of gravity:

$$f_s = mg \sin \theta, \quad f_{s,\max} = \mu mg \cos \theta.$$

The block is at rest without  $F$  whenever  $f_s \leq f_{s,\max}$ , i.e.

$$\boxed{\mu \geq \tan \theta}.$$

**Extremum 2: the block does not move even with  $F \rightarrow \infty$ .** With  $F$  applied, the normal force grows:

$$N = mg \cos \theta + F \sin \theta \implies f_{s,\max} = \mu mg \cos \theta + \mu F \sin \theta.$$

At the slip limit the up-slope *component of  $F$*  equals the friction plus the down-slope component of the weight,

$$F \cos \theta - mg \sin \theta - f_s = 0 \implies f_s = F \cos \theta - mg \sin \theta \leq f_{s,\max},$$

so

$$F \cos \theta - mg \sin \theta \leq \mu mg \cos \theta + \mu F \sin \theta \implies \mu \geq \frac{F \cos \theta - mg \sin \theta}{mg \cos \theta + F \sin \theta}.$$

Taking  $F \rightarrow \infty$ ,

$$\lim_{F \rightarrow \infty} \frac{F \cos \theta - mg \sin \theta}{mg \cos \theta + F \sin \theta} = \cot \theta,$$

so no finite  $F$  can move the block when

$$\boxed{\tan \theta \geq \frac{1}{\mu}}.$$

**The range of  $F$  in the three regimes.**

The two extrema partition the  $(\theta, \mu)$  parameter space into three cases. The definite formulas (when they exist) are obtained from  $F_{\text{net}} = 0$  at the two slip limits.

For  $F_{\min}$ , friction acts up-slope at its maximum:

$$F \cos \theta + \mu (mg \cos \theta + F \sin \theta) = mg \sin \theta \implies \boxed{F_{\min} = mg \frac{\sin \theta - \mu \cos \theta}{\cos \theta + \mu \sin \theta} = mg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}.$$

For  $F_{\max}$ , friction acts down-slope at its maximum:

$$\mu (mg \cos \theta + F \sin \theta) = F \cos \theta - mg \sin \theta \implies \boxed{F_{\max} = mg \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} = mg \frac{\tan \theta + \mu}{1 - \mu \tan \theta}}.$$

Hence:

1. If  $\mu \geq \tan \theta$ : the block stays put with no force, so

$$\boxed{F_{\min} = 0}, \quad \boxed{F_{\max} = mg \frac{\tan \theta + \mu}{1 - \mu \tan \theta}}.$$

2. If  $\tan \theta \geq \frac{1}{\mu}$ : no finite force moves the block, so

$$\boxed{F_{\min} = mg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}, \quad \boxed{F_{\max} \rightarrow \infty}.$$

3. If  $\mu \leq \tan \theta \leq \frac{1}{\mu}$ : both bounds are finite,

$$\boxed{F_{\min} = mg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}, \quad \boxed{F_{\max} = mg \frac{\tan \theta + \mu}{1 - \mu \tan \theta}}.$$